Resolution Geometry: The Physical Path

The Stress Symbol, Kinetic Realizability, Remainder-Driven Defect Conservation, and the Gravitational No-Go

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Abstract

The gravitation computation imports a metric (operational route) or selects a field equation (dynamical route); neither is RG computing physics. We ask what RG computes *itself*, from its second-order layer and its final remainder, and we attack every candidate lock. Three results lock. (i) *The stress symbol.* The stress tensor is the raw second-moment natural module of the momentum law, $T = \int p \otimes p \, d\mu$; its block form relative to an observer is exactly RG's second-order datum with $p = 1$, $q = 3$; and for the thermal law it is $Z(g_F + \mu \otimes \mu)$, the raw Fisher second moment of the thermal family in the inverse-temperature covector. (ii) *Kinetic realizability.* A symmetric tensor is the stress of some positive momentum law on the causal cone iff it is positive semidefinite on covectors and has nonnegative η-trace; every such tensor is the stress of four future causal beams of equal invariant weight (constructive proof). The weak, null, strong and dominant energy conditions are then theorems, not declarations; the cone is strictly smaller than the dominant energy condition (non-null electromagnetic fields, the stiff fluid and Λ are excluded with certificates); kinetic stress is Hawking–Ellis type I unless it is pure null dust, and the Landau frame is the observer that minimizes measured energy density. (iii) *Remainder-driven dynamics.* Principal-stress defects carry RG's wall charge; their total charge is conserved under *every* continuous evolution and they are created or annihilated only in opposite $\pm\frac{1}{2}$ pairs, independently of any constitutive law or field equation. One result is a proved *no-go*: no function of the source remainder determines the gravitational field (Minkowski and Schwarzschild share $T = 0$; Nordström and Einstein vacua share $T = 0$ and differ in light bending). The Weyl part is the gravitational remainder; its Petrov types are multiplicity types, locked for diagonalizable types and open for Jordan types. Conservation $\nabla_a T^{ab} = 0$ follows from declared free-fall transport, not from the remainder. Verdict: RG computes the source side and a topological law of motion; it does not compute gravitational dynamics, which remain a selection or a measurement. Battery: 15 attacks (11 locked, 1 refuted, 2 boundary, 1 no-go), 5 canaries rejected, master seed 20261023.

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1 The question and the three modes

The gravitation computation [3] is honest about its two entry routes: a metric is reconstructed from anchored clocks and light (operational), or a field equation is declared and solved (dynamical). The ladder then selects Einstein's equation by Lovelock's theorem, in the same way that it selects Fisher by Čencov's [1]. Both are *selections*. The question of this paper is sharper: *does RG compute physics from its own objects, and in particular from its final remainder?*

We separate three modes and keep them separate throughout.

(M1) **Selection.** A physical law is chosen among declared alternatives by a characterization theorem (Čencov, Lovelock) or by data. RG certifies the choice; it does not produce the law.

(M2) **Computation.** A physical task is computed from declared observation laws by RG constructions (natural modules, quotients, remainder), with a certificate, and with no physics-specific dynamical law imported.

(M3) **Constraint.** A conserved quantity or selection rule is imposed on *every* admissible dynamics by the remainder, without fixing an equation of motion.

The canonical connection $B = PJ^\top(JPJ^\top)^{-1}$ is the one existing case in which RG supplies an equation of motion (under H-resp) [4]. This paper adds computations (M2) and a constraint (M3) on the source side, and proves that the gravitational side admits neither.

Result	Mode	Status
Stress symbol, block map, Fisher–stress identity	M2	locked (Thms 3.1, 3.2)
Kinetic cone, four-beam decomposition	M2	locked (Thm 4.3)
Energy conditions	M2	forced, not declared (Cor 4.4)
Hawking–Ellis type, Landau frame	M2	locked (Thm 4.6)
Source law from T	M2	refuted: correspondence (Prop 4.7)
Defect charge conservation	M3	locked (Thm 6.1)
$\nabla_a T^{ab} = 0$	M1	conditional on declared transport (Prop 7.1)
Gravitational field from source remainder	—	no-go (Thm 8.1)
Petrov types as remainder types	M2	locked for I, D, O; open for II, III, N

Table 1: What locks.

2 Setting

Let V be a four-dimensional real vector space with Lorentzian form η of signature $(+,-,-,-)$, a time orientation, and future causal cone $L^+ = \{p : \ p \cdot p \geq 0, \ p^0 \geq 0\}$. Indices are raised and lowered with η; T^{ab} is contravariant; $\eta(T) = \eta_{ab}T^{ab}$.

Definition 2.1 (Momentum protocol)**.** A *momentum protocol* at an event records particle four-momenta. Its law is a finite positive Borel measure μ on L^+ with finite second moments (number weighting included). A finite law is $\mu = \sum_i w_i \delta_{p_i}$.

Definition 2.2 (Stress symbol)**.** $T_\mu = \int_{L^+} p \otimes p \, d\mu(p) \in \mathrm{Sym}^2 V$, and $N_\mu = \int p \, d\mu$.

3 The stress symbol in RG

Theorem 3.1 (Symbol and block map)**.** (i) *$\mu \mapsto T_\mu$ is a natural second-order module of the momentum law: for every Lorentz map Λ, $T_{\Lambda_\sharp \mu} = \Lambda T_\mu \Lambda^\top$, and $\eta(T)$ is invariant.*

(ii) *For an observer u with simultaneity $u^\perp$, the blocks $T = \left(\begin{smallmatrix} \rho & j \\ j^\top & S \end{smallmatrix}\right)$ are RG's second-order datum (Σ_R, C, Σ_N) with $R = \mathbb{R}u$ $(p = 1)$ and $N = u^\perp$ $(q = 3)$. If T is positive definite, the canonical correlation $\sigma = \|S^{-1/2} j\| / \sqrt{\rho}$ satisfies $\sigma < 1$.*

(iii) *The coupling j is observer-relative, in the same way that RG's coupling is complement-relative [2]. It vanishes exactly in a frame whose u is an eigenvector of $T^a{}_b$ (the Landau frame, Theorem 4.6).*

Proof. (i) $\int (\Lambda p) \otimes (\Lambda p) \, d\mu = \Lambda T \Lambda^\top$, and η is Λ-invariant. (ii) The block decomposition is that of a covariance on $R \oplus N$; $\sigma < 1$ is equivalent to positivity of the Schur complement $\rho - j S^{-1} j^\top$. (iii) $j = 0$ iff $T^{ab} u_b \propto u^a$. □

Theorem 3.2 (Fisher–stress identity)**.** *Let Π be the invariant measure on a mass shell, β a future timelike covector, and $Z(\beta) = \int e^{-\beta_a p^a} d\Pi(p)$. For the thermal law $d\mu_\beta = e^{-\beta \cdot p} d\Pi$,*

$$T^{ab}_{\mu_\beta} = \frac{\partial^2 Z}{\partial \beta_a \partial \beta_b} = Z(\beta)\big(g_F^{ab}(\beta) + m^a m^b\big), \qquad g_F = \partial^2 \log Z, \qquad m = N/Z,$$

where g_F is the Fisher metric of the normalized thermal family $\{\mu_\beta / Z(\beta)\}$ in its natural parameter $\beta = u/\Theta$. g_F is positive definite.

Proof. Differentiate under the integral: $\partial_a Z = -N^a$, $\partial_a \partial_b Z = T^{ab}$. For an exponential family, $\partial^2 \log Z$ is the covariance of the sufficient statistic p, which is the Fisher metric; $\partial^2 \log Z = T/Z - N \otimes N / Z^2$. Positivity: the covariance of p is positive definite unless the law is concentrated on a hyperplane. □

Remark 3.3. This is the symbol bridge asked for. RG's Fisher module and the physical stress tensor are the centred and raw second moments of the same thermal law; the Lorentzian form enters only through the mass shell. This is consistent with the positive-cost obstruction of [3]: the Fisher metric on β-space is Riemannian, and η is not a congruence of it. The identity is the Massieu-function form of relativistic thermodynamics [7]; the RG reading is new only as a reading.

4 Kinetic realizability

Definition 4.1. $\mathcal{K} = \{T \in \mathrm{Sym}^2 V : \ T(\xi,\xi) \geq 0 \ \forall \xi \in V^*, \ \eta(T) \geq 0\}$.

Both conditions are invariant under every linear congruence and under Lorentz maps respectively, so $\mathcal{K}$ is a Lorentz-invariant closed convex cone.

Lemma 4.2 (Constant diagonal)**.** *Every real symmetric $k \times k$ matrix M is orthogonally similar to a matrix whose diagonal entries all equal* $\operatorname{tr} M/k$.

Proof. Let $a = \operatorname{tr} M/k$. If some diagonal entry differs from a, there are i, j with $M_{ii} > a > M_{jj}$. The (i,i) entry of $G_\theta^\top M G_\theta$, for the rotation G_θ in the (i,j) plane, varies continuously from M_{ii} at $\theta = 0$ to M_{jj} at $\theta = \pi/2$, so it equals a for some θ. Other diagonal entries except the (j,j) one are unchanged and the trace is preserved. Each step fixes one more entry at a; after at most $k-1$ steps all are equal [8]. □

Theorem 4.3 (Kinetic cone)**.** $\{T_\mu\} = \mathcal{K}$. *Moreover every $T \in \mathcal{K}$ is the stress of exactly four future causal beams of equal invariant weight:*

$$T = \sum_{l=1}^{4} p_l \otimes p_l, \qquad p_l \in L^+, \qquad p_l \cdot p_l = \tfrac{1}{4}\eta(T).$$

Proof. ($\subseteq$) $T_\mu(\xi,\xi) = \int (\xi \cdot p)^2 d\mu \geq 0$ and $\eta(T_\mu) = \int p \cdot p \, d\mu \geq 0$. ($\supseteq$) Write $T = VV^\top$ with V a real 4×4 matrix. The matrix $M = V^\top \eta V$ is symmetric with $\operatorname{tr} M = \eta(T) \geq 0$. Lemma 4.2 gives an orthogonal Q with $(Q^\top M Q)_{ll} = \eta(T)/4$. Put $p_l = VQe_l$. Then $\sum_l p_l \otimes p_l = VQQ^\top V^\top = T$ and $p_l \cdot p_l = (Q^\top M Q)_{ll} \geq 0$, so each p_l is causal; replace p_l by $-p_l$ if it is past-directed, which leaves $p_l \otimes p_l$ unchanged. □

Corollary 4.4 (Energy conditions are forced)**.** *For $T \in \mathcal{K}$, unit future timelike u and null k:*

$$T_{ab}u^a u^b \geq 0, \quad T_{ab}k^a k^b \geq 0, \quad \big(T_{ab} - \tfrac{1}{2}\eta(T)\eta_{ab}\big)u^a u^b \geq 0, \quad -T^a{}_b u^b \text{ future causal.}$$

Proof. With the beams of Theorem 4.3: $T(u,u) = \sum (p_l \cdot u)^2$; $T(k,k) = \sum (p_l \cdot k)^2$; by the reversed Cauchy–Schwarz inequality $(p \cdot u)^2 \geq p \cdot p$, so $T(u,u) - \frac{1}{2}\eta(T) = \sum [(p_l \cdot u)^2 - \frac{1}{2} p_l \cdot p_l] \geq 0$; and $T^a{}_b u^b = \sum p_l^a (p_l \cdot u)$ is a nonnegative combination of future causal vectors. □

Proposition 4.5 (Strictness and the fluid bound)**.** (a) *A perfect fluid* $\mathrm{diag}(\rho, p, p, p)$ *is in $\mathcal{K}$ iff $\rho \geq 0$ and $0 \leq p \leq \rho/3$. An anisotropic Landau-frame tensor* $\mathrm{diag}(\rho, p_1, p_2, p_3)$ *is in $\mathcal{K}$ iff $p_i \geq 0$ and $\sum p_i \leq \rho$.*

(b) *The stiff fluid ($\eta(T) = -2\rho$), the cosmological constant (negative pressure) and every non-null electromagnetic field (a tension $T(\xi,\xi) < 0$ along the field) lie outside $\mathcal{K}$. Each certificate is a covector ξ with $T(\xi,\xi) < 0$ or the sign of $\eta(T)$. Null electromagnetic fields lie in $\mathcal{K}$ (photon beams).*

(c) *The electric field and the stiff fluid satisfy the dominant energy condition. Hence $\mathcal{K}$ is strictly smaller than the dominant energy cone.*

Proof. (a) Positivity is a congruence invariant, so it can be checked in the rest frame; the trace condition is $\rho - \sum p_i \geq 0$. (b) For $E \parallel e_1$, $T^{11} = -E^2/2$; a non-null field has a frame in which $E \parallel B$ and the same tension appears. (c) Standard. □

Theorem 4.6 (Hawking–Ellis type and the Landau frame)**.** *Let $0 \neq T = T_\mu$.*

(i) *If* $\operatorname{supp} \mu$ *is not contained in a single null ray, then $E(u) = T(u^\flat, u^\flat)$ is coercive on the future unit hyperboloid. It attains its minimum $\rho_L > 0$ at a unique u_L, and u_L is the unique timelike eigenvector of $T^a{}_b$ (type I).*

(ii) *If* $\operatorname{supp}\mu$ *lies in one null ray* $\mathbb{R}_+k$*, then* $T = c\,k\otimes k$*,* $\inf E = 0$ *is not attained, and* $T^a{}_b$ *has no timelike eigenvector (type II, null dust).*

Proof. Write $u = \gamma(1, v)$ with $|v| < 1$. Then $E(u) = \gamma^2 F(v)$ with $F(v) = \int (p^0 - \vec{p}\cdot v)^2 d\mu$, which is continuous on the closed unit ball. $F(v) = 0$ at $|v| = 1$ iff μ is supported on the null ray along $(1, v)$. Under (i), $F \geq c > 0$ on the closed ball, so $E \geq c\gamma^2 \to \infty$, and E attains a minimum. The Lagrange condition for E on $\{u \cdot u = 1\}$ is $T^a{}_b u^b = \rho_L u^a$. Uniqueness: two timelike eigenvectors with distinct eigenvalues would be η-orthogonal, which is impossible. If a common eigenspace contained a timelike u and a unit spacelike x, then $T(x^\flat, x^\flat) = \rho_L x \cdot x < 0$, contradicting $T \in \mathcal{K}$. (ii) $E(u) = c(k \cdot u)^2 > 0$ and tends to 0 along boosts toward k; the eigenvectors of $T^a{}_b = c\,k^a k_b$ are k and $k^\perp$, which contains no timelike vector. □

Proposition 4.7 (The source remains a correspondence)**.** *T does not determine μ: a 400-particle gas and its four-beam decomposition have the same T but different third moments. The source fibre over T is a CORRESPONDENCE, refined by a third-moment protocol. The trace $\eta(T) = \int p \cdot p\, d\mu$ fixes the total invariant mass weight.*

5 The remainder of stress fields

Proposition 5.1 (Multiplicities control isotropy)**.** *For type I stress $T = \operatorname{diag}(\rho, p_1, p_2, p_3)$ in its Landau frame, with principal pressures of multiplicities m_j, the Lorentz stabilizer is $S\big(\prod_j \mathrm{O}(m_j)\big) \subset \mathrm{SO}(3)$ whenever no $p_i = -\rho$. The perfect fluid has dimension 3, the axisymmetric case 1, the anisotropic case 0. Dust has Lorentz stabilizer dimension 3; Λ, which lies outside $\mathcal{K}$, has 6.*

Proof. A Lorentz stabilizer fixes the unique timelike eigenvector (Theorem 4.6), so it lies in the rotations of $u_L^\perp$, where it commutes with S. The dimensions are computed exactly in the battery (P08). □

Theorem 5.2 (Remainder of stress fields)**.** *Let T be a type I stress field on a manifold U whose principal-pressure multiplicities are locally constant (H-type). Then:*

(i) *The Landau frame u_L is a smooth, globally time-oriented unit field.*

(ii) *The eigenbundles $E_j \subset u_L^\perp$ of the principal pressures are smooth Euclidean subbundles.*

(iii) *Up to isometries over id_U that intertwine T, the field is determined by its local moduli (ρ, p_j) and the isomorphism classes of (E_j). This is the Global Structure Theorem of [2], with its proof via Riesz projections, for a single self-adjoint operator.*

In the free contract (morphisms may change frames), the class of a rank-one block over a loop is $w_1 \in \mathbb{Z}_2$. In the ambient-fixed contract (frames of $u_L^\perp$ declared), it is a homotopy class of loops in the projective space. In two dimensions $[S^1, \mathbb{R}P^1] = \mathbb{Z}$, so the index of a line field is half an integer.

6 Remainder-driven dynamics: defect charge

Let $\Omega \subset \mathbb{R}^2$ be a domain and $\sigma : \Omega \times [0,1] \to \mathrm{Sym}^2\mathbb{R}^2$ a continuous evolving plane stress field. Put $D = \frac{1}{2}(\sigma_{11} - \sigma_{22}) + i\sigma_{12}$. The isotropic points at time t are $Z_t = D(\cdot, t)^{-1}(0)$, and the principal direction is $\theta = \frac{1}{2}\arg D$.

Theorem 6.1 (Defect charge conservation)**.** (i) *For a loop $\gamma \subset \Omega$ that avoids Z_t, the charge $q_\gamma(t) = \frac{1}{2}\deg(D(\cdot, t) \circ \gamma) \in \frac{1}{2}\mathbb{Z}$ is the rotation number of the principal line field. Around a single defect, the principal direction returns with sign $(-1)^{2q}$: this is the $\mathbb{Z}_2$ wall charge of [2].*

(ii) *$q_\gamma(t)$ is constant on every time interval during which γ avoids Z_t. If Z_t is finite inside γ, then $q_\gamma = \sum_z q_z$.*

(iii) *At a transversal zero, $q_z = \frac{1}{2}\operatorname{sign}\det D_x D$. In a generic one-parameter evolution, defects are created or annihilated only in pairs of opposite charge $\pm\frac{1}{2}$.*

None of this uses a constitutive law, an equation of motion or a field equation.

Proof. (i) $\arg D = 2\theta$, so the winding of D is twice the rotation of θ; a half-turn of θ reverses the direction vector. (ii) The degree is a homotopy invariant of maps $S^1 \to \mathbb{C}\setminus 0$, and it is additive over a decomposition into small loops around the zeros [9]. (iii) The local degree of a regular zero is the sign of the Jacobian. For generic families, $Z = \{(x,t) : D = 0\}$ is a one-manifold in $\Omega \times [0,1]$. Its points of tangency to time slices are folds, where $\det D_x D$ changes sign, so the two zeros meeting there have opposite charge. Charge is conserved there by (ii). □

Example 6.2 (An exact elastic witness; P09, P10)**.** The Airy stress function $\phi = \operatorname{Re}\big(\bar{z}z^3/6 - tz^2/2 + i\varepsilon z^3/6\big)$ is biharmonic. The stress $\sigma_{11} = \phi_{yy}$, $\sigma_{22} = \phi_{xx}$, $\sigma_{12} = -\phi_{xy}$ satisfies equilibrium and compatibility exactly, and

$$D = t - |z|^2 + \varepsilon y + i\varepsilon x.$$

The isotropic points are $(0, \frac{1}{2}(\varepsilon \pm \sqrt{\varepsilon^2 + 4t}))$, with charges $+\frac{1}{2}$ and $-\frac{1}{2}$. They merge at $t_c = -\varepsilon^2/4$ and annihilate. The charge on any fixed enclosing loop is 0 before and after. Seeded random families confirm that every generic defect has charge $\pm\frac{1}{2}$, and that every change of a loop charge coincides with a zero crossing the loop (P10b).

Remark 6.3 (Physical content and limits)*.* Isotropic points are observable in photoelastic fringe patterns, and their pairwise appearance and disappearance under changing load is a testable prediction of Theorem 6.1; no real photoelastic data were used here. In three dimensions, coincidences of two principal stresses have codimension two. They form lines, and the two colliding eigenbundles carry $\mathbb{Z}_2$ monodromy around them by the forced wall monodromy theorem of [2]; for spacetime stress fields the corresponding loci are worldsheets. The tensor-topology literature [10, 11] describes the same singularities. RG's contribution is to derive them as the remainder of a declared second-order observation and to type their status.

7 Conservation is a declared transport

Proposition 7.1 (Conservation from free-fall transport)**.** *If the momentum law at each event is transported by the geodesic flow of a declared metric (Vlasov transport), then $\nabla_a T^{ab} = 0$ [5]. The battery checks the flat 1+1 case: the relative divergence is 2.6×10^{-9} for free streaming and 0.55 for a non-conservative decay (P14). Conservation is therefore a consequence of the declared transport (free fall is a physical coupling hypothesis), not of the remainder.*

8 The gravitational no-go

Theorem 8.1 (No source-remainder gravity)**.** *No map from source data (the stress field, or the full momentum law) on a region to the curvature on that region is consistent with general relativity, nor with any theory admitting several vacua.*

(i) *Minkowski space and the Schwarzschild exterior both have $T = 0$ and $\mu = 0$. The first has Kretschmann scalar 0; the second has Ricci $= 0$ and Kretschmann scalar $48m^2/r^6$.*

(ii) *Nordström's theory has the vacuum $g = \varphi^2\eta$ with $\Delta\varphi = 0$, $\varphi = 1 - m/r$. It also has $T = 0$ outside the source. It is conformally flat, so it bends no light, whereas general relativity bends light by $\alpha = 0.1427$ at $r_0 = 30m$.*

Proof. (i) is computed symbolically (P12). (ii) Null geodesics are conformally invariant, and φ is harmonic. □

Remark 8.2 (What this means). A field equation algebraic in the Ricci tensor determines at most Ricci, which has 10 components. The 10 Weyl components are free; they are the gravitational remainder. Once Einstein's equation is selected, the Weyl field evolves by the Bianchi identity, with source the derivative of T. That evolution is forced *given* the selection, and not before it. Theory choice between Einstein and Nordström is an RG CORRESPONDENCE that data resolve: light bending discriminates, as in the frozen predictions of [3].

Proposition 8.3 (Petrov types as remainder types; P13)**.** *Let $Q = E + iB$ be the Weyl operator on the complex three-space of self-dual bivectors, acted on by* $\mathrm{SO}(3,\mathbb{C}) \cong \mathrm{SO}^+(1,3)$.

(i) *For diagonalizable Q, the multiplicity type of the eigenvalues determines the isotropy. Type I has real dimension* 0*, type D (Schwarzschild,* $\propto \mathrm{diag}(-2,1,1)$*) has dimension* 2 *(a boost and a rotation), and type O has dimension* 6*.*

(ii) *Types II, III and N are not diagonalizable (type N: $Q^2 = 0 \neq Q$). They fall outside the self-adjoint remainder theorem. Their classification as remainder data is open.*

9 The physical path ladder

Status codes are those of the print edition: F forced, I imported, H hypothesis-conditioned, N no-go, O open.

Rung	Statement	Status	Witness
Φ0	Momentum protocol; law on the causal cone	input	—
Φ1	Stress symbol is a natural second-order module; block map to (Σ_R, C, Σ_N) with $p = 1, q = 3$	F	P01, P08
Φ2	Kinetic cone $\mathcal{K}$; four equal-weight beams	F	P02, P03
Φ3	WEC, NEC, SEC, DEC forced	F	P05
Φ4	$\mathcal{K} \subsetneq$ DEC; $0 \leq p \leq \rho/3$; non-null EM, Λ, stiff excluded	F / N	P04
Φ5	Hawking–Ellis I or II; variational Landau frame	F	P06
Φ6	T does not determine the source law	N	P07
Φ7	Fisher–stress identity for the thermal family	F	P16
Φ8	Multiplicities control isotropy	F	P08
Φ9	Remainder of stress fields: moduli plus eigenbundles; $\mathbb{Z}_2$ (free) versus $\frac{1}{2}\mathbb{Z}$ (ambient-fixed)	F (H-type)	Thm 5.2
Φ10	Defect charge conservation; opposite-pair creation and annihilation	F	P09, P10, P10b
Φ11	$\nabla_a T^{ab} = 0$ from declared free-fall transport	H: transport	P14
Φ12	No source-remainder gravity; Weyl is the gravitational remainder	N	P12
Φ13	Field equation by selection (Lovelock) or by data (light bending)	I / H	[3]
Φ14	Petrov I, D, O as multiplicity types	F	P13
Φ15	Petrov II, III, N; stratified gravitational remainder	O	P13
Φ16	Field (non-kinetic) source protocols, e.g. electromagnetic tension	O	P04

10 Battery

`rg_physical_path_attacks.py` (NumPy, SciPy, SymPy; master seed 20261023, per-attack seed $+101k$) runs 15 attacks and 5 canaries; a manifest with the source hash is written first. Exact arithmetic is used for every algebraic claim: Airy equilibrium and compatibility, the formula for D, stabilizer dimensions, the Schwarzschild Ricci tensor and Kretschmann scalar, and harmonicity of the Nordström potential. Seeded floating point is used for cones, observers, transport and quadrature. The decisive quantities are signs, integers or residuals below 10^{-7}.

The canaries are five false statements, all rejected:

- a non-null EM field is kinetic;
- the index of an isotropic point is an integer;
- $T = 0$ forces flat space;
- null dust has a rest frame;
- perturbed beams still reproduce T.

Limits. The battery is not blind. All fixtures are analytic or synthetic, and no photoelastic, astrophysical or laboratory data are used. Numerical minimization and quadrature support, but do not replace, the proofs.

11 Non-claims and the answer

1. No gravitational dynamics is derived from RG. Theorem 8.1 proves that the source remainder cannot supply it.
2. No claim is made that all physical stress is kinetic. Field stresses lie outside $\mathcal{K}$ and need field protocols (Φ16).
3. No novelty is claimed for the classical results used: energy conditions of kinetic matter, the Hawking–Ellis classification [6], the Massieu-function identity, degree theory, tensor-field topology, Petrov types.
4. The defect law is a constraint on all dynamics, not an equation of motion.

Answer. RG computes physics in two modes, M2 and M3. On the source side it computes the stress symbol, its realizability, the energy conditions, the rest frame, the isotropy and the remainder. As a law of motion it supplies a conservation law for stress defects that every dynamics must obey. It does not compute the gravitational field: the remainder of the source leaves the Weyl curvature free, so the field equation stays a selection or a measurement. The next lock that would change this verdict is an RG-native observation protocol whose remainder *contains* the Weyl data, for example tidal or geodesic-deviation protocols, together with a proof that its dynamics are forced. That is open.

References

[1] M. Rada, *Resolution Geometry: Indistinguishability, Its Symmetries, and the Geometry of What Observations Determine*, print edition, 2026.

[2] M. Rada, *The Final Remainder of Resolution Geometry*, 2026.

[3] M. Rada, *Resolution Geometry: A Conditional Gravitation Computation*, 2026.

[4] M. Rada, *Physical Curvature: The Canonical Connection as a Law of Motion*, 2026.

[5] J. Ehlers, General relativity and kinetic theory, in *General Relativity and Cosmology* (Proc. Int. School Enrico Fermi XLVII), Academic Press, 1971.

[6] S. W. Hawking, G. F. R. Ellis, *The Large Scale Structure of Space-Time*, Cambridge, 1973.

[7] W. Israel, Nonstationary irreversible thermodynamics: a causal relativistic theory, *Ann. Phys.* 100 (1976), 310–331.

[8] R. A. Horn, C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge, 2013.

[9] J. Milnor, *Topology from the Differentiable Viewpoint*, University Press of Virginia, 1965.

[10] T. Delmarcelle, L. Hesselink, The topology of symmetric, second-order tensor fields, *Proc. IEEE Visualization* 1994, 140–147.

[11] J. F. Nye, *Natural Focusing and Fine Structure of Light*, IOP Publishing, 1999.